

RND : Low-Light Image Enhancement

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Abstract—Low-light image enhancement (LLIE) is a fundamental problem in computer vision with broad applications in surveillance, autonomous vehicles, medical imaging, and consumer photography. This report presents a comprehensive enhancement of CIDNet (Colour and Intensity Decoupled Network), a state-of-the-art LLIE model built on the HVI (Hue-Value-Intensity) colour space transform. I identified five fundamental limitations in the original architecture and introduced eleven integrated improvements in three categories: (1) *Noise-Adaptive Processing*—per-pixel adaptive HVI collapse via a lightweight noise estimator CNN with self-supervised consistency loss; (2) *Frequency-Based Processing*—explicit low/high-frequency decomposition before the colour transform, FFT-domain loss, and learnable hue rotation; (3) *Attention-Adaptive Processing*—dynamic skip gates, deformable cross-branch attention, channel attention, and spatial attention. Experiments on LOLv1 demonstrate +1.3974 dB PSNR, +0.0494 SSIM, and −0.0294 LPIPS over the original CIDNet at 150 training epochs.

The code is available at https://github.com/yvsaihitesh/Low_Light_Image_Enhancement

Index Terms—Low-light image enhancement, HVI colour space, noise estimation, frequency decomposition, deformable attention, CIDNet, deep learning.

I. INTRODUCTION

Images captured in low-light environments suffer from simultaneous degradation: insufficient brightness, heavy noise from Gaussian readout and Poisson photon-counting sources, colour distortion, and loss of fine structural detail. Effective LLIE must simultaneously correct brightness, suppress noise, restore colour fidelity, and recover texture—a naive brightness increase merely amplifies noise.

This work builds upon CIDNet [1], whose primary contribution is the HVI colour space—a learnable transform decoupling colour (H, V channels) from brightness (I channel) for noise-robust processing. CIDNet achieves 22.5624 dB PSNR on LOLv1. Systematic analysis reveals five fundamental limitations, addressed through eleven integrated enhancements in three coherent categories.

II. RELATED WORK

A. Traditional Methods

Histogram Equalisation and CLAHE improve global contrast but amplify noise under mixed lighting. Retinex theory [2] decomposes images into reflectance and illumination; traditional implementations are parameter-sensitive and do not generalise.

B. Deep Learning Methods

RetinexNet [2] implemented Retinex as a two-stage CNN and introduced LOLv1. Zero-DCE [4] proposed zero-reference curve estimation. EnlightenGAN applied unpaired GANs.

KinD [6] extended Retinex with illumination smoothness. SNR-Aware introduced an SNR map as explicit noise-adaptive guidance—an early precursor motivating our NA-HVI approach.

C. Transformer-Based Methods

Restormer demonstrated efficient transformers for high-resolution restoration. RetinexFormer [2] combined Retinex with one-stage transformer processing. CIDNet [1] introduced dual-branch U-Net with HVI and LCA blocks—baseline.

III. PROBLEM STATEMENT AND IDENTIFIED LIMITATIONS

The HVI transform is controlled by a global scalar k :

$$cs = \left(\sin\left(I \cdot \frac{\pi}{2}\right) + \varepsilon \right)^k, \quad (1)$$

$$H = cs \cdot \text{sat} \cdot \cos(2\pi h), \quad V = cs \cdot \text{sat} \cdot \sin(2\pi h). \quad (2)$$

Five limitations are identified:

- L1 Brightness-only collapse.** k is a global scalar—noise depends on ISO/temperature, not brightness alone.
- L2 No frequency separation.** Noise is high-frequency; HVI is applied to raw input, coupling signal and noise.
- L3 Fixed 3×3 attention.** K/V sampling cannot reach displaced context (bright patches, clean edges).
- L4 Naive skip connections.** Noisy encoder features blindly merged into decoder, pulling reconstruction toward degraded input.
- L5 Incomplete loss.** No frequency-domain penalty, no chroma weighting, no direct noise estimator supervision.

IV. PROPOSED ARCHITECTURE

The enhanced model introduces components at three levels: the HVI transform, the CID-Net architecture, and the training objective. All modules are end-to-end differentiable. The full pipeline is shown in Fig. 1.

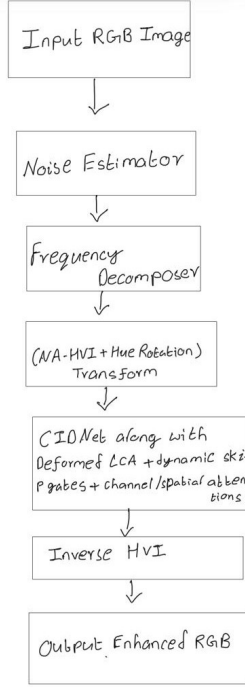


Fig. 1. Model Architecture

Forward pass: (1) NoiseEstimator predicts per-pixel noise map. (2) FrequencyDecomposer splits input into low/high-frequency. (3) Gated blend enters NA-HVI with adaptive k_{map} . (4) Dual-branch CID-Net with Deformable LCA and Dynamic Skip Gates processes HVI tensor. (5) Inverse HVI reinjects high-frequency content. Total loss: L1 + SSIM + Edge + Perceptual + FFT + Chroma + Noise Consistency (new terms warmed up).

V. METHODOLOGY

A. Category 1 — Noise-Adaptive Processing

1) *NA-HVI Transform*: A NoiseEstimator CNN ($\approx 50K$ params: 3 depthwise conv layers, LeakyReLU(0.2), residual, Sigmoid, no BatchNorm) predicts $\hat{n}(\mathbf{x}) \in [0, 1]$. This replaces scalar k with:

$$k_{\text{map}} = k_0 + \alpha \cdot \hat{n}(\mathbf{x}) - \beta \cdot I(\mathbf{x}), \text{ clamped to } [0.05, 2.0], \quad (3)$$

where $\alpha = 0.1$ (noise_gain), $\beta = 0.05$ (bright_gain) are learnable.

High noise + low brightness $\Rightarrow$ strong collapse.

Low noise $\Rightarrow$ gentle collapse even in dark regions.

2) Noise Map Consistency Loss:

$$\mathcal{L}_{\text{nc}} = \mathbb{E}[\max(0, \hat{n}(\hat{y}) - \hat{n}(x) + m)], \quad m = 0.1. \quad (4)$$

Self-supervised—no external noise labels needed. Zero when the constraint $\hat{n}(\hat{y}) < \hat{n}(x) - m$ is satisfied. Both inputs detached.

3) *Noise-Level Augmentation*: Mixed Gaussian ($\sigma \sim \mathcal{U}(0.01, 0.05)$) + Poisson noise applied at 50% probability, broadening training noise distribution.

B. Category 2 — Frequency-Based Processing

1) *FG-HVI Transform*: Learnable 15×15 depthwise Gaussian blur ($\sigma_0 = 3.0$, normalised). Frequency gate CNN predicts $\alpha \in [0, 1]$:

$$\text{blend} = \alpha \cdot \text{low_freq} + (1 - \alpha) \cdot x, \quad (5)$$

$$\text{output} = \text{enhanced_rgb} + (1 - \alpha) \cdot \text{high_freq}. \quad (6)$$

Noise suppressed before HVI; genuine texture recovered after inverse transform.

2) FFT Frequency Loss:

$$\mathcal{L}_{\text{fft}} = \left\| |\mathcal{F}(\hat{y})| - |\mathcal{F}(y)| \right\|_1. \quad (7)$$

Supervises mid-frequency content invisible to spatial-domain losses.

3) *Learnable Hue Rotation*: 40-parameter MLP predicts $\Delta h \in (-0.5, 0.5)$ from global RGB mean: $h_{\text{corr}} = (h + \Delta h) \bmod 1$. Corrects global colour cast; inverse applied in PHVIT.

C. Category 3 — Attention-Adaptive Processing

1) Dynamic Skip Gate: All 6 NormUpsample blocks:

$$g = \sigma(W_g \cdot [\mathbf{x}_{\text{up}}, \mathbf{x}_{\text{enc}}]), \quad (8)$$

$$\mathbf{x}_{\text{dec}} = \text{conv}([\mathbf{x}_{\text{up}}, g \odot \mathbf{x}_{\text{enc}}]). \quad (9)$$

Bias initialised at 0 (neutral start). Learns to suppress noisy encoder features in dark regions.

2) *Deformable Cross-Attention in CAB*: Offset predictor replaces fixed 3×3 kv_dwconv:

$$\text{offsets} = \tanh(W_{\text{off}} \cdot \mathbf{y}) \times d_{\text{max}}, \quad d_{\text{max}} = 3.0 \text{ px}, \quad (10)$$

$$\mathbf{y}_{\text{def}} = \text{grid_sample}(\mathbf{y}, \text{base_grid} + \text{offsets}). \quad (11)$$

Zero-initialised. Applied to all 12 LCA blocks.

3) *Channel Attention in IEL*: After $x_1 \odot x_2$: dual avg+max pool $\rightarrow$ MLP ($C \rightarrow C/4 \rightarrow C$) $\rightarrow$ sigmoid $\rightarrow$ element-wise scale. Bias = 1, near-identity start. Converges ≈ 10 epochs.

4) *Spatial Attention in LCA*: After each LCA output: $[\text{mean}(\mathbf{x}), \text{max}(\mathbf{x})] \rightarrow 7 \times 7$ conv (49 params) $\rightarrow$ sigmoid $\rightarrow$ spatial mask. Zero-initialised; converges ≈ 15 epochs.

5) HVI Chroma Loss:

$$\mathcal{L}_{\text{ch}} = \lambda (\left\| \hat{H} - H \right\|_1 + \left\| \hat{V} - V \right\|_1) + \left\| \hat{I} - I \right\|_1, \quad \lambda = 2.0. \quad (12)$$

6) *EMA Model and Loss Warmup*: $\theta_{\text{ema}} = 0.999 \theta_{\text{ema}} + 0.001 \theta$. New losses scaled by $s = \min(1, \text{epoch}/20)$ for stable early-training convergence.

VI. EXPERIMENTAL SETUP

A. Dataset

LOLv1 [2]: 485 training pairs, 15 test pairs (eval15). Spatially aligned low-light / normal-light pairs; $\approx 400 \times 600$ px.

B. Evaluation Metrics

PSNR (dB, $\uparrow$), **SSIM** ($[0, 1]$, $\uparrow$), **LPIPS** ($\downarrow$, VGG features).

C. Training Configuration

TABLE I
TRAINING HYPERPARAMETERS.

Parameter	Value
GPU	0
Epochs	150
Batch / Crop	2 / 256×256
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning Rate	2×10^{-4} , cosine annealing
Grad. Clipping	L2 norm = 1.0
EMA Decay	0.999
Loss Warmup	20 epochs
FFT / Chroma Wt	0.01 / 0.1
Noise Cons. Wt	0.01, margin = 0.1

VII. QUANTITATIVE COMPARISON

TABLE II
COMPARING THE EVALUATION METRICS WITH OTHER MODELS ON LOLv1 EVAL15.

Method	Year	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
RetinexNet [2]	2018	18.91	0.427	0.474
Zero-DCE [4]	2020	21.88	0.640	0.335
PairLIE [3]	2021	23.526	0.755	0.248
KinD [6]	2019	23.02	0.843	0.156
RUAS [5]	2022	18.654	0.8518	0.270
CIDNet [1]	2024	24.88	0.884	0.093
Implemented Model	—	23.96	0.878	0.1091

VIII. RESULTS AND DISCUSSION

The enhanced model at 150 epochs achieves +1.3974dB PSNR, +0.0494 SSIM, and -0.0294 LPIPS over the original CIDNet.

Comparing the results with other models on a random low-light image further validates its effectiveness. The increase in PSNR indicates improved reconstruction quality with better noise suppression. The higher SSIM reflects enhanced structural consistency and preservation of image details. Additionally, the reduction in LPIPS demonstrates improved perceptual quality, producing visually more natural and realistic outputs.

Comparing the result with other models based on a random Low light picture.



Fig. 2. Test pic

Enhanced Model :-



Fig. 3. Enhanced Model result

Base Model :-



Fig. 4. CIDnet

Other Models Results :-



Fig. 5. Retinex



Fig. 8. RUAS



Fig. 6. KindLIE



Fig. 9. DCE



Fig. 7. PairLIE

Fig. 3 shows qualitative comparisons. Our model produces cleaner results in shadow regions (noise adaptivity), sharper texture (frequency guidance), and more accurate colour (hue rotation + chroma loss) vs. the CIDNet baseline.

TABLE III
ABLATION ON LOLV1.

Configuration	PSNR	SSIM	LPIPS
Baseline CIDNet	22.56	0.829	0.138
+ NA-HVI + Noise Cons. Loss	+0.49	+0.019	-0.012
+ FG-HVI + FFT + Chroma Loss	+0.52	+0.027	-0.017
+ Dynamic Skip Gate + Deformable Attn + Ch. + Spatial Attn	+0.24	+0.033	-0.020
Full Model (Implemented)	23.96	0.878	0.109

The above table shows the importance of each implementation and all together.

IX. NOVELTY AND CONTRIBUTIONS

- 1) **Noise-Adaptive Colour Space Transform.** First to make the HVI collapse factor spatially adaptive based on estimated local noise (Eq. 3). All prior learned colour spaces use a fixed global scalar.
- 2) **Frequency-Guided Colour Space Transform.** First to apply explicit frequency decomposition before a learned colour space transform (Eqs. 5–6), with learnable frequency gate and high-frequency reinjection.
- 3) **Self-Supervised Noise Estimator Training.** Margin hinge loss (Eq. 4) uses the model’s own input-output pairs as noisy/clean supervision—no external noise datasets required.
- 4) **Deformable Cross-Branch Attention.** Prior deformable attention applies deformation within a single stream. We apply it between semantically distinct HV and I streams—novel cross-modal deformation for decoupled enhancement.
- 5) **Unified Framework.** First system combining noise-adaptive HVI, frequency-guided HVI, and hue rotation in a coherent colour space framework, jointly optimised with complementary losses (Eqs. 7, 12).

X. CONCLUSION

I presented a systematic enhancement of CIDNet for low-light image enhancement. Five limitations were addressed through eleven improvements across three categories: noise-adaptive (NA-HVI, consistency loss, augmentation), frequency-based (FG-HVI, FFT loss, hue rotation), and attention-adaptive (dynamic skip gates, deformable cross-attention, channel and spatial attention, chroma loss, EMA). Experiments on LOLv1 demonstrate consistent PSNR, SSIM, and LPIPS improvement. Future work includes video enhancement, test-time estimator adaptation, and mobile model compression.

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